

Urban Economics

Urban Sprawl and Land-Use Controls

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Outline for Today

1.



Next week

Readings



Introduction





Urban sprawl has been drawing strong feelings

Critics say:

- Increases traffic congestion and pollution 🚗
- Consumes too much land 🌲
- Less incentives to develop downtown areas 🏢
- Low density sprawl reduces social interaction 👤👤
- Dependence on automobiles increases obesity rates

The data shows:

- Spatial growth was really high---from 1.3% to 1.9% of land area got developed from 1976 to 1992 (Burchfield et al., 2006) 📈
- Other sources show that the rate is even higher

As a result, many cities have implemented land-use controls to limit sprawl



Growth management policies to control urban sprawl

At least 12 states have growth management policies

New Jersey committed in 1998 to spend \$1 billion to purchase vacant land to limit



Do the critics have a point?

Do cities take up too much space?

Should they implement land-use controls to limit sprawl?

The Empirical Evidence



This table shows the results from two papers

Elasticities of urban area land with respect to various variables.

	Brueckner and Fansler (1983)	McGrath (2005)
Population (L)	1.10	0.76
Commuting cost (t)	0	-0.28
Income (y)	1.50	0.33
Agricultural rent (r_A)	-0.23	-0.10

The results tells us if X increases by 1 percent, Y increases by how much percent?

Both studies show that higher population leads to more land use

- 1 percent increase in population leads to 1.1 and 0.76 percent increase in land use

Income effects are large: 1 percent increase in income leads to 1.5 and 0.3 percent increase in land use

Higher agricultural rent leads to less land use, mixed evidence on the effect of commuting costs

It is easy to see why urban sprawl happens in the US

So, if we are growing in population and getting richer, is it bad that we are sprawling out?

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So, if we are growing in population and getting richer, is it bad that we are sprawling out?

Well, it depends

If this sprawl is being driven by consumer preferences, then it is **not a problem**

However, if sprawl is being driven by market failures, then it is a **problem**

Market failures related present in economic process that determines size of cities

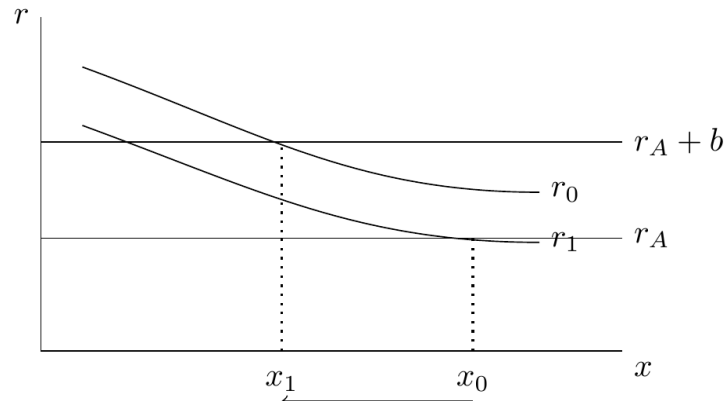
1. The amenity value of open space

- Assume each acre of open space provides b dollars above and beyond its agricultural value
- This benefit is not captured when land is converted from agricultural to urban use
- A landowner can either rent the land to a developer r rent it out to a farmer r_a
- If $r > r_a$, the land will be developed
- Society would ideally want the landowner to take into account the amenity value b
- So society would want the land developed only if $r + b > r_a \Rightarrow$ city boundary rent should be $r = r_a - b$

Socially optimal effect agricultural rent

1. The amenity value of open space

- So a optimal city size would emerge if the rent is $r = r_a - b$, which includes the amenity value of open space



- How do we get from x_0 to x_1 ?
 - By implementing land-use controls that limit development
 - Imposing a development tax

This model assumes that people value open space and the government knows the value

Is this realistic?

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Is this realistic?

It is not hard to imagine that the person designing anti-sprawl policies might reach a value that is larger than that of the average person

This could lead to reducing dwelling sizes and make living in the city more costly (people do not value the open space that is saved)

So land-use controls could lead to a loss in welfare if the value of open space is overestimated



Market failures related present in economic process that determines size of cities

2. The market failure related to traffic congestion



Market failures related present in economic process that determines size of cities

2. The market failure related to traffic congestion

Why congestion is a market failure

- Each additional car on a congested road **slows down all other drivers**
- Drivers bear only their **private commuting cost**, not the **external delay** they cause
- Result: The **private cost < social cost**, leading to **too much driving** and **excessively large cities**

The Congestion Externality

- The *externality* arises because a single extra car increases travel time for all others
- Each driver's additional time loss is small, but **summed across all drivers**, the total cost is **non-negligible**
- No individual driver has the incentive to account for this social cost
- Thus:
 - Commuting decisions are distorted
 - City size is **too large** in the free-market equilibrium

The Effect on Urban Spatial Structure

Without congestion tolls:

- Commuting is **too cheap**
- Residents are willing to live farther away
- The equilibrium city radius would be **larger than optimal**

With congestion tolls:

- Higher commuting cost → shorter optimal commutes
- Population locates **closer to the center**
- The city becomes **more compact**

Congestion Tolls as a Policy Tool

What the toll should equal:

- A congestion toll must equal the **external cost** a driver imposes on others
- When this toll is charged:
 - Commuters internalize the full social cost
 - Traffic volumes fall
 - The **city shrinks to the efficient size**

Key insight:

Tolls make the market mimic the social optimum.

Evidence from Real-World Congestion Pricing

Cities implementing congestion pricing:

- **Singapore** (one of the earliest and most comprehensive systems)
- **London** (central-city cordon charge)
- **Stockholm** (peak-hour bridge tolls)
- **New York City**

What studies find:

- After toll adoption, cities become **more compact**
- Residents choose **shorter commuting distances**
- Urban models correctly predict **declines in city size** when tolls are introduced

From the News

The New York Times

New York Today

The Streets Are Safer This Year for Drivers, Cyclists and Pedestrians

There have been 87 traffic deaths in the city so far this year, the lowest number since 2018. Officials credit the Vision Zero program and congestion pricing.

Local News

Congestion pricing reducing traffic in NYC, increasing revenue after first 6 months, officials say

By Jesse Zanger

Welfare Effects of Correcting Congestion Externalities

- By increasing the cost of commuting, tolls:
 - Reduce excessive driving
 - Align **private** and **social** commuting costs
 - Lead to **higher welfare** than the no-toll equilibrium
- Since congestion makes free-market cities **too large**,
charging congestion tolls moves the city toward the socially optimal spatial size.